

Vector Calculus Notes: Green's Theorem & Surface Integrals

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1 Green's Theorem in the Plane

Statement

Let R be a closed, bounded region in the xy -plane whose boundary C consists of finitely many smooth curves. Let $P(x, y)$ and $Q(x, y)$ be functions that are continuous and have continuous partial derivatives $\frac{\partial P}{\partial y}$ and $\frac{\partial Q}{\partial x}$ everywhere in the domain. Then:

$$\iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \oint_C P dx + Q dy = \oint_C \vec{F}(\vec{r}) \cdot d\vec{r}$$

where $\vec{F} = [P, Q]$ and $\vec{r} = (x, y)$.

Area Calculation Using Green's Theorem

Setting combinations of P and Q :

Case 1: Setting $Q = x$ and $P = 0$:

$$\text{Area} = \iint_R dx dy = \oint_C x dy$$

Case 2: Setting $Q = 0$ and $P = -y$:

$$\text{Area} = \iint_R dx dy = - \oint_C y dx$$

Case 3: Setting $Q = x$ and $P = -y$:

$$\begin{aligned} \iint_R 2 dx dy &= \oint_C x dy - y dx \\ \implies \text{Area} &= \frac{1}{2} \oint_C (x dy - y dx) \end{aligned}$$

Area in Polar Coordinates

Using transformations $x = r \cos \theta$ and $y = r \sin \theta$:

$$dx = \frac{\partial x}{\partial r} dr + \frac{\partial x}{\partial \theta} d\theta = \cos \theta dr - r \sin \theta d\theta$$

$$dy = \frac{\partial y}{\partial r} dr + \frac{\partial y}{\partial \theta} d\theta = \sin \theta dr + r \cos \theta d\theta$$

Substituting into the area formula:

$$\begin{aligned} \text{Area} &= \frac{1}{2} \oint_C (x dy - y dx) \\ &= \frac{1}{2} \int \left[r \cos \theta (\sin \theta dr + r \cos \theta d\theta) - r \sin \theta (\cos \theta dr - r \sin \theta d\theta) \right] \\ &= \frac{1}{2} \int r^2 (\cos^2 \theta + \sin^2 \theta) d\theta \\ &= \frac{1}{2} \int r^2 d\theta \end{aligned}$$

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Examples of Green's Theorem

Example 1: Using Green's theorem, evaluate the line integral $\oint_C \vec{F}(\vec{r}) \cdot d\vec{r}$, where $\vec{F} = [\ln \cosh 2x, 2y^2 \sinh y]$ and C is the boundary of the region.

Here, $P(x, y) = \ln \cosh 2x$ and $Q(x, y) = 2y^2 \sinh 2y$.

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0 - 0 = 0$$

$$\oint_C \vec{F}(\vec{r}) \cdot d\vec{r} = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \iint_R 0 dx dy = 0$$

Example 2: Using Green's theorem, evaluate the line integral $\oint_C \vec{F}(\vec{r}) \cdot d\vec{r}$, where $\vec{F} = [x \cosh 2y, 2x^2 \sinh 2y]$ and C is the boundary of the region $x^2 \leq y \leq x$.

Here, $P(x, y) = x \cosh 2y$ and $Q(x, y) = 2x^2 \sinh 2y$.

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 4x \sinh 2y - 2x \sinh 2y = 2x \sinh 2y$$

Region limits: $x^2 \leq y \leq x \implies x^2 = x \implies x(x - 1) = 0 \implies 0 < x < 1$.

$$\begin{aligned}
\oint_C \vec{F}(\vec{r}) \cdot d\vec{r} &= \int_0^1 \int_{x^2}^x 2x \sinh 2y \, dy \, dx \\
&= \int_0^1 [x \cosh 2y]_{x^2}^x \, dx \\
&= \int_0^1 (x \cosh 2x - x \cosh 2x^2) \, dx \\
&= \left[\frac{1}{2} x \sinh 2x - \frac{1}{4} \cosh 2x - \frac{1}{4} \sinh 2x^2 \right]_0^1 \\
&= \left(\frac{1}{2} \sinh 2 - \frac{1}{4} \cosh 2 - \frac{1}{4} \sinh 2 \right) - \left(0 - \frac{1}{4} - 0 \right) \\
&= \frac{1}{4} (1 + \sinh 2 - \cosh 2)
\end{aligned}$$

Example 3: Using Green's theorem, evaluate the line integral $\oint_C \vec{F}(\vec{r}) \cdot d\vec{r}$ where $\vec{F} = [y, 2x]$ and $C : \frac{x^2}{4} + \frac{y^2}{16} = 1$.

$$\begin{aligned}
\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} &= 2 - 1 = 1 \\
\oint_C \vec{F}(\vec{r}) \cdot d\vec{r} &= \iint_R 1 \, dx \, dy = \text{Area of Ellipse} = \pi \cdot a \cdot b = \pi(2)(4) = 8\pi
\end{aligned}$$

Example 4: Find the area of the region under the cardioid $r = 1 + 2 \cos \theta$ for $0 < \theta < \frac{\pi}{2}$.

$$\begin{aligned}
\text{Area} &= \frac{1}{2} \int_0^{\pi/2} r^2 \, d\theta = \frac{1}{2} \int_0^{\pi/2} (1 + 2 \cos \theta)^2 \, d\theta \\
&= \frac{1}{2} \int_0^{\pi/2} (1 + 4 \cos \theta + 4 \cos^2 \theta) \, d\theta \\
&= \frac{1}{2} \int_0^{\pi/2} [3 + 4 \cos \theta + 2 \cos 2\theta] \, d\theta \\
&= \frac{1}{2} [3\theta + 4 \sin \theta + \sin 2\theta]_0^{\pi/2} \\
&= \frac{1}{2} \left[\frac{3\pi}{2} + 4 \right] = \frac{3\pi}{4} + 2
\end{aligned}$$

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2 Parametrization of 3D Surfaces

Surfaces in 3D space can be represented in multiple forms:

- **Implicit:** $f(x, y, z) = c$
- **Explicit:** $z = f(x, y)$
- **Parametric:** $\vec{r}(u, v) = [f(u, v), g(u, v), h(u, v)]$ for $a \leq u \leq b$, $c \leq v \leq d$

Standard Surface Parametrizations

1. **Cone** ($x^2 + y^2 = z^2$):

$$\vec{r}(u, v) = (u \cos v, u \sin v, u), \quad 0 \leq v \leq 2\pi, \quad 0 \leq u \leq a$$

2. **Sphere** ($x^2 + y^2 + z^2 = a^2$):

$$\vec{r}(u, v) = (a \cos u \sin v, a \sin u \sin v, a \cos v), \quad 0 \leq u \leq 2\pi, \quad 0 \leq v \leq \pi$$

3. **Cylinder** ($x^2 + y^2 = a^2, z = b$):

$$\vec{r}(u, v) = (a \cos v, a \sin v, u), \quad 0 \leq v \leq 2\pi, \quad c \leq u \leq d$$

3 Surface Area & Surface Integrals

Surface Area Formula (Parametric)

The surface area A of a parametric surface $\vec{r}(u, v)$ over region R is:

$$A = \iint_S dA = \iint_R |\vec{r}_u \times \vec{r}_v| du dv$$

Example: Surface Area of a Sphere of radius a Using parametric form $\vec{r}(u, v)$ for a sphere:

$$\begin{aligned} |\vec{r}_u \times \vec{r}_v| &= a^2 \sin u \\ A &= \int_0^{2\pi} \int_0^\pi a^2 \sin u du dv = a^2 \int_0^{2\pi} [-\cos u]_0^\pi dv = a^2(2)(2\pi) = 4\pi a^2 \end{aligned}$$

Surface Area Formula (Explicit $z = f(u, v)$)

For an explicit surface $z = f(u, v)$, parametrized as $\vec{r} = (u, v, f(u, v))$:

$$\begin{aligned} \vec{r}_u &= (1, 0, f_u), \quad \vec{r}_v = (0, 1, f_v) \\ \vec{r}_u \times \vec{r}_v &= (-f_u, -f_v, 1) \implies |\vec{r}_u \times \vec{r}_v| = \sqrt{1 + f_u^2 + f_v^2} \\ A &= \iint_R \sqrt{1 + f_u^2 + f_v^2} du dv \end{aligned}$$

Surface Integrals of Vector Fields (Flux)

The flux of a vector field $\vec{F}$ through a surface S is:

$$\iint_S \vec{F} \cdot \hat{n} dA = \iint_R \vec{F}(\vec{r}(u, v)) \cdot (\vec{r}_u \times \vec{r}_v) du dv$$

where $\vec{N} = \vec{r}_u \times \vec{r}_v$ is the normal vector to the surface.

Example 1: Evaluate $\iint_S \vec{F} \cdot \hat{n} dA$ where $\vec{F} = [x^2, y^2, z^2]$ and S is parametrized by $\vec{r}(u, v) = [u \cos v, u \sin v, 3v]$ for $0 \leq u \leq 1$ and $0 \leq v \leq 2\pi$.

1. Compute derivatives:

$$\vec{r}_u = [\cos v, \sin v, 0]$$

$$\vec{r}_v = [-u \sin v, u \cos v, 3]$$

2. Compute normal vector $\vec{N} = \vec{r}_u \times \vec{r}_v$:

$$\vec{N} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos v & \sin v & 0 \\ -u \sin v & u \cos v & 3 \end{vmatrix} = [3 \sin v, -3 \cos v, u]$$

3. Substitute into field $\vec{F}(\vec{r}(u, v)) = [u^2 \cos^2 v, u^2 \sin^2 v, 9v^2]$:

$$\vec{F} \cdot \vec{N} = 3u^2 \cos^2 v \sin v - 3u^2 \sin^2 v \cos v + 9uv^2$$

4. Evaluate double integral:

$$\begin{aligned} \iint_S \vec{F} \cdot \hat{n} dA &= \int_0^{2\pi} \int_0^1 (3u^2 \cos^2 v \sin v - 3u^2 \sin^2 v \cos v + 9uv^2) du dv \\ &= \int_0^{2\pi} \left[u^3 \cos^2 v \sin v - u^3 \sin^2 v \cos v + \frac{9}{2} u^2 v^2 \right]_0^1 dv \\ &= \int_0^{2\pi} \left(\cos^2 v \sin v - \sin^2 v \cos v + \frac{9}{2} v^2 \right) dv \\ &= \left[-\frac{1}{3} \cos^3 v - \frac{1}{3} \sin^3 v + \frac{3}{2} v^3 \right]_0^{2\pi} \\ &= \left(-\frac{1}{3} - 0 + 12\pi^3 \right) - \left(-\frac{1}{3} - 0 + 0 \right) = 12\pi^3 \end{aligned}$$

Example 2: Evaluate $\iint_S G(\vec{r}) dA$ where $G(\vec{r}) = (x^2 + y^2)^2 = z^2$ and S is given by $\vec{r}(u, v) = (u \cos v, u \sin v, 2u)$ for $0 \leq u \leq 1$, $-\pi \leq v \leq \pi$.

1. Compute partial derivatives:

$$\vec{r}_u = (\cos v, \sin v, 2), \quad \vec{r}_v = (-u \sin v, u \cos v, 0)$$

2. Compute cross product $\vec{r}_u \times \vec{r}_v$:

$$\vec{r}_u \times \vec{r}_v = (-2u \cos v, -2u \sin v, u)$$

3. Compute magnitude $|\vec{r}_u \times \vec{r}_v|$:

$$|\vec{r}_u \times \vec{r}_v| = \sqrt{(-2u \cos v)^2 + (-2u \sin v)^2 + u^2} = \sqrt{4u^2 + u^2} = \sqrt{5}u$$

4. Express $G(\vec{r})$ in parametric form:

$$G = (u^2 \cos^2 v + u^2 \sin^2 v)^2 - (2u)^2 = u^4 - 4u^2$$

5. Evaluate integral:

$$\begin{aligned}\iint_S G(\vec{r}) dA &= \int_{-\pi}^{\pi} \int_0^1 (u^4 - 4u^2) \sqrt{5} u du dv \\&= \sqrt{5} \int_{-\pi}^{\pi} \left[\int_0^1 (u^5 - 4u^3) du \right] dv \\&= \sqrt{5} \int_{-\pi}^{\pi} \left[\frac{u^6}{6} - u^4 \right]_0^1 dv \\&= \sqrt{5} \int_{-\pi}^{\pi} \left(\frac{1}{6} - 1 \right) dv = \sqrt{5} \left(-\frac{5}{6} \right) (2\pi) = -\frac{5\sqrt{5}\pi}{3}\end{aligned}$$

4 Surface Integrals and Flux (Page 1)

Problem Statement

Evaluate the surface integral $\iint_S \vec{F} \cdot \hat{n} dA$ where:

$$\vec{F} = [x^2, e^y, 1]$$

over the surface $S : x + y + z = 1$ in the first octant ($x \geq 0, y \geq 0, z \geq 0$).

Parametrization and Region Limits

Let $x = u$ and $y = v$, then $z = 1 - u - v$.

Since $z \geq 0$:

$$1 - u - v \geq 0 \implies v \leq 1 - u$$

Given $y \geq 0 \implies v \geq 0$, we have:

$$0 \leq v \leq 1 - u$$

Since $x \geq 0 \implies u \geq 0$, it follows that $0 \leq u \leq 1$.

Tangent and Normal Vectors

The position vector is $\vec{r}(u, v) = [u, v, 1 - u - v]$.

$$\vec{r}_u = [1, 0, -1]$$

$$\vec{r}_v = [0, 1, -1]$$

The normal vector $\hat{n}$ is:

$$\hat{n} = \vec{r}_u \times \vec{r}_v = [1, 1, 1]$$

Evaluating the Integration

The vector field in terms of parameters u and v :

$$\vec{F} = [u^2, e^v, 1]$$

Dot product $\vec{F} \cdot \hat{n}$:

$$\vec{F} \cdot \hat{n} = [u^2, e^v, 1] \cdot [1, 1, 1] = u^2 + e^v + 1$$

Now, evaluate the double integral:

$$\begin{aligned} \iint_S \vec{F} \cdot \hat{n} dA &= \int_0^1 \int_0^{1-u} (u^2 + e^v + 1) dv du \\ &= \int_0^1 [u^2 v + e^v + v]_0^{1-u} du \\ &= \int_0^1 (u^2(1-u) + e^{1-u} + 1 - u - (0 + e^0 + 0)) du \\ &= \int_0^1 (-u^3 + u^2 - u + e^{1-u}) du \\ &= \left[-\frac{u^4}{4} + \frac{u^3}{3} - \frac{u^2}{2} - e^{1-u} \right]_0^1 \\ &= \left(-\frac{1}{4} + \frac{1}{3} - \frac{1}{2} - 1 \right) - (0 + 0 - 0 - e) \\ &= -\frac{1}{4} + \frac{1}{3} - \frac{1}{2} - 1 + e \\ &= e - \frac{17}{12} \end{aligned}$$

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5 Volume Integrals and Mass (Page 2)

General Definitions

$$\text{Volume Integral} = \iiint_V dx dy dz$$

$$\text{Mass} = \iiint_V \rho dx dy dz$$

Example 1: Mass of a Region

Find the total mass if the density is $\rho = 4z$ in a region of the first octant bounded by $y = 1 - x^2$ and $z = x$.

Limits of Integration

* In the first octant ($x \geq 0, y \geq 0, z \geq 0$): * From $y = 1 - x^2 \geq 0 \implies x^2 \leq 1 \implies 0 \leq x \leq 1$. * Limits for y : $0 \leq y \leq 1 - x^2$. * Limits for z : $0 \leq z \leq x$.

Integral Computation

$$\begin{aligned}\text{Mass} &= \int_0^1 \int_0^{1-x^2} \int_0^x 4z \, dz \, dy \, dx \\&= \int_0^1 \int_0^{1-x^2} [2z^2]_0^x \, dy \, dx \\&= \int_0^1 \int_0^{1-x^2} 2x^2 \, dy \, dx \\&= 2 \int_0^1 x^2 [y]_0^{1-x^2} \, dx \\&= 2 \int_0^1 x^2(1-x^2) \, dx \\&= 2 \int_0^1 (x^2 - x^4) \, dx \\&= 2 \left[\frac{x^3}{3} - \frac{x^5}{5} \right]_0^1 \\&= 2 \left(\frac{1}{3} - \frac{1}{5} \right) = 2 \left(\frac{2}{15} \right) = \frac{4}{15}\end{aligned}$$

Example 2: Volume Integral over a Tetrahedron

Find the volume integral where $\rho = 4x + y + \pi \sin(\pi z)$, bounded by the tetrahedron with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$.

Boundary plane equation: $x + y + z = 1 \implies z = 1 - x - y \geq 0$. Limits: $0 \leq z \leq 1 - x - y$, $0 \leq y \leq 1 - x$, and $0 \leq x \leq 1$.

6 Volume Integrals Continued (Page 3)

Tetrahedron Integral Set-Up

$$\int_0^1 \int_0^{1-x} \int_0^{1-x-y} (4x + y + \pi \sin(\pi z)) \, dz \, dy \, dx$$

Example 3: Volume Enclosed by an Ellipsoid

Find the volume enclosed by the surface $S : 4x^2 + y^2 + 9z^2 = 36$.

Limits Derivation

$$\begin{aligned}4x^2 + y^2 + 9z^2 &\leq 36 \\9z^2 &\leq 36 - 4x^2 - y^2 \implies -\sqrt{36 - 4x^2 - y^2} \leq 3z \leq \sqrt{36 - 4x^2 - y^2} \\-\frac{1}{3}\sqrt{36 - 4x^2 - y^2} &\leq z \leq \frac{1}{3}\sqrt{36 - 4x^2 - y^2}\end{aligned}$$

For y :

$$36 - 4x^2 - y^2 \geq 0 \implies y^2 \leq 36 - 4x^2 \\ -\sqrt{36 - 4x^2} \leq y \leq \sqrt{36 - 4x^2}$$

For x :

$$36 - 4x^2 \geq 0 \implies 4x^2 \leq 36 \implies x^2 \leq 9 \implies -3 \leq x \leq 3$$

7 Stokes' Theorem & Curl (Page 4)

Stokes' Theorem Formula

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\text{curl } \vec{F}) \cdot \hat{n} dS$$

Curl Formula

$$\text{curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P(x, y, z) & Q(x, y, z) & 0 \end{vmatrix} = \hat{i}(0 - 0) - \hat{j}(0 - 0) + \hat{k} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right)$$

Example Problem

Evaluate over $S : z = y^2$ where $0 \leq x \leq 1$ and $0 \leq y \leq 2$. Let:

$$\vec{F} = [e^z, e^z \sin y, e^z \cos y]$$

1. Compute Curl of $\vec{F}$

$$\text{curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ e^z & e^z \sin y & e^z \cos y \end{vmatrix}$$

$$\begin{aligned} \text{curl } \vec{F} &= \hat{i} \left(\frac{\partial}{\partial y}(e^z \cos y) - \frac{\partial}{\partial z}(e^z \sin y) \right) - \hat{j} \left(\frac{\partial}{\partial x}(e^z \cos y) - \frac{\partial}{\partial z}(e^z) \right) + \hat{k}(\dots) \\ &= \hat{i}(-e^z \sin y - e^z \sin y) - \hat{j}(0 - e^z) + \hat{k}(0 - 0) \\ &= -2e^z \sin y \hat{i} + e^z \hat{j} \end{aligned}$$

2. Surface Parametrization

Let $x = u$ and $y = v$, then $z = v^2$.

$$\vec{r}(u, v) = (u, v, v^2)$$

$$\vec{r}_u = (1, 0, 0) = \hat{i}$$

$$\vec{r}_v = (0, 1, 2v) = \hat{j} + 2v\hat{k}$$

$$\hat{n} = \vec{r}_u \times \vec{r}_v = \hat{i} \times (\hat{j} + 2v\hat{k}) = -2v\hat{j} + \hat{k}$$

8 Stokes' Theorem Solution (Page 5)

Parametric Form of Curl

Substituting $z = v^2$ into the curl expression:

$$\text{curl } \vec{F}(u, v) = -2e^{v^2} \sin v \hat{i} + e^{v^2} \hat{j}$$

Dot Product with Normal Vector

$$(\text{curl } \vec{F}) \cdot \hat{n} = \left(-2e^{v^2} \sin v \hat{i} + e^{v^2} \hat{j} \right) \cdot (-2v \hat{j} + \hat{k}) = -2ve^{v^2}$$

Surface Integral Computation

Limits: $0 \leq u \leq 4, 0 \leq v \leq 2$.

$$\begin{aligned} \iint_S (\text{curl } \vec{F}) \cdot \hat{n} dA &= \int_0^2 \int_0^4 -2ve^{v^2} du dv \\ &= -2 \int_0^2 ve^{v^2} [u]_0^4 dv \\ &= -4 \int_0^2 2ve^{v^2} dv \\ &= -4 \left[e^{v^2} \right]_0^2 \\ &= -4(e^4 - e^0) = 4(1 - e^4) \end{aligned}$$

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Line Integral Verification along Segment OA

Along segment OA on the boundary: $y = 0, z = 0, x = t$ where $0 \leq t \leq 4$.

Position vector and differential:

$$\vec{r}(t) = (t, 0, 0) \implies \frac{d\vec{r}}{dt} = (1, 0, 0)$$

Vector field evaluated on OA :

$$\vec{F}(\vec{r}(t)) = [e^0, e^0 \sin 0, e^0 \cos 0] = (1, 0, 1)$$

$$\vec{F}(\vec{r}(t)) \cdot \frac{d\vec{r}}{dt} = (1, 0, 1) \cdot (1, 0, 0) = 1$$

Line integral along OA :

$$\int_{OA} \vec{F} \cdot d\vec{r} = \int_0^4 1 dt = 4$$